

behaviour of a PWA can be similar to that of a native app. Push alerts and background sync are two other characteristics that must be considered. A responsive design framework must use pre-cache and lazy loading to ensure rapid loading.

Progressive Web of things

The progressive Web of things is designed to allow IoT protocols to connect to devices. Browsers can execute these programs, and are capable of displaying these apps on any device. Mobile phones, desktop computers, laptop computers, televisions and smartwatches are all examples of devices. Geolocation, Bluetooth, push notifications, camera, and NFC are all elements of HTML5 Web APIs. For the progressive Web of things, APIs linked to Web USB, WebRTC and Web NFC are critical. Web RTC is a technology that allows video and audio to be streamed between two devices.

“With PWA studio, merchants can deliver app-like mobile commerce that is fast, integrated and reliable. These experiences are proven to improve SEO and to drive 50 per cent higher conversion rates, and 50 per cent better page performance on average.”

— Magento executive

Application, networking, data transmission, and sensing are all layers in a typical IoT architecture. PWA apps are in the application layer and use the data communication and sensing layers to provide functionality. The energy, media, healthcare, and telecom

industries all use these applications. Progressive Web apps can help with remote device administration, which is an important feature.

Device connection, varied technology stacks, and many application protocols are all issues with progressive Web of things apps. Smart home apps use pre-caching and data sync features to provide data. These apps can work without a network connection, but they require it to access real-time data.

Ionic framework, electron framework, Node.JS, and other JavaScript frameworks are used to create progressive Web of things apps. The sensing layer of the IoT architecture contains IoT sensors and actuators. The networking layers include routers and gateways.

Discoverability of the progressive Web of things

Devices may be accessed using the progressive Web of things application via HTTP and Web sockets. These programs address design elements such as the device's URL/resources, payload information, request semantics and responses.

W3C manifests and service worker registration can be used to find the devices. The devices can also be found via search engines. The resources will be linked from the websites.

The discovery component of the Web of Things design ensures interoperability. This component aids in the discovery of gadgets. Metadata can be shared across devices to aid in discovery. URL, name, current location, and status are all part of the metadata. Sensors, actuators, commands, and attributes are all part of the device sensing metadata. mDNS, lightweight data models, Web item models, and JSON-LD are some of the network discovery protocols used for device discovery.

Security and IoT safety

Security is a major design consideration. With the Internet of Things and PWA apps, a connected world opens up a slew of possibilities. The main concern is security. Credit cards, personal information, sensitive information, and passwords are all security concerns. If these issues emerge from a PWA website, trust will be lost. Device security is critical in the progressive Web of things to ensure spam messages are not delivered.

Attacks on the Internet of Things are prevalent and harmful. Information regarding a person's running time, medicines taken and other personal details is sensitive. Security is an issue for automobiles when it comes to physical devices. Clear text passwords, faulty certificates, outdated program versions, and other flaws might all be a problem.

For secrecy, integrity, and identity, Web content on the progressive Web of things must be safeguarded. If the website is collecting private and sensitive information about the user, it must be secure. Snooping, denial of service, phishing, and security breaches are all possibilities.

HTTPS helps to secure the website while having no effect on its performance. HTTP/2 is significantly quicker than HTTP. HTTP/2 is used by HTTPS-enabled websites. If the PWA website is not secure, some browsers may issue a warning. This is one of the reasons why the progressive Web of things should be HTTPS enabled.

Performance

The user experience and performance of a progressive Web of things app must be improved. Responsive design makes it easier to work with a variety of devices and form factors. Service workers are used to address offline conditions.